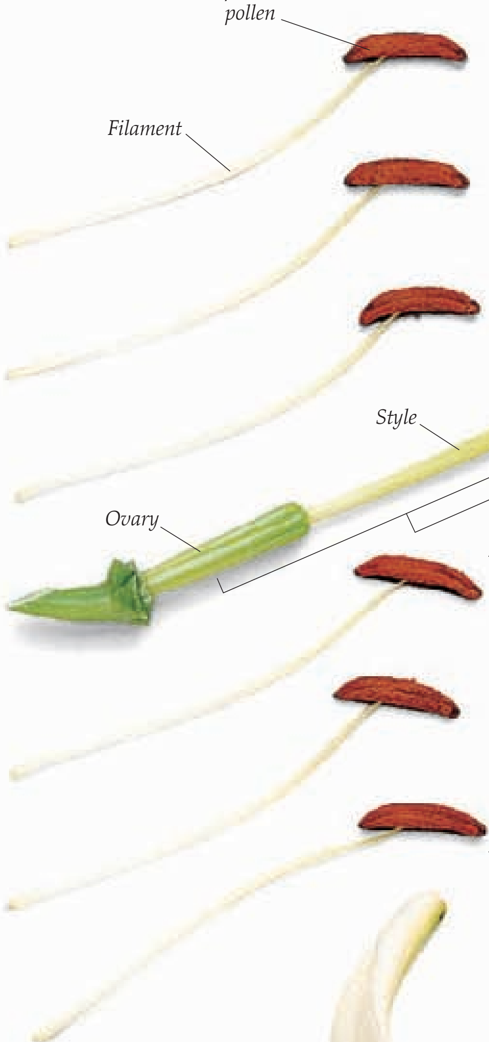
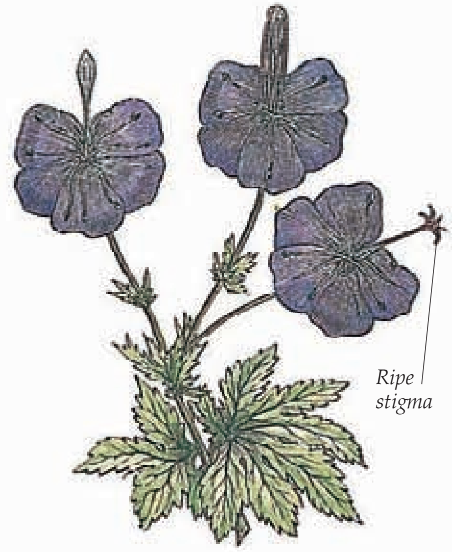


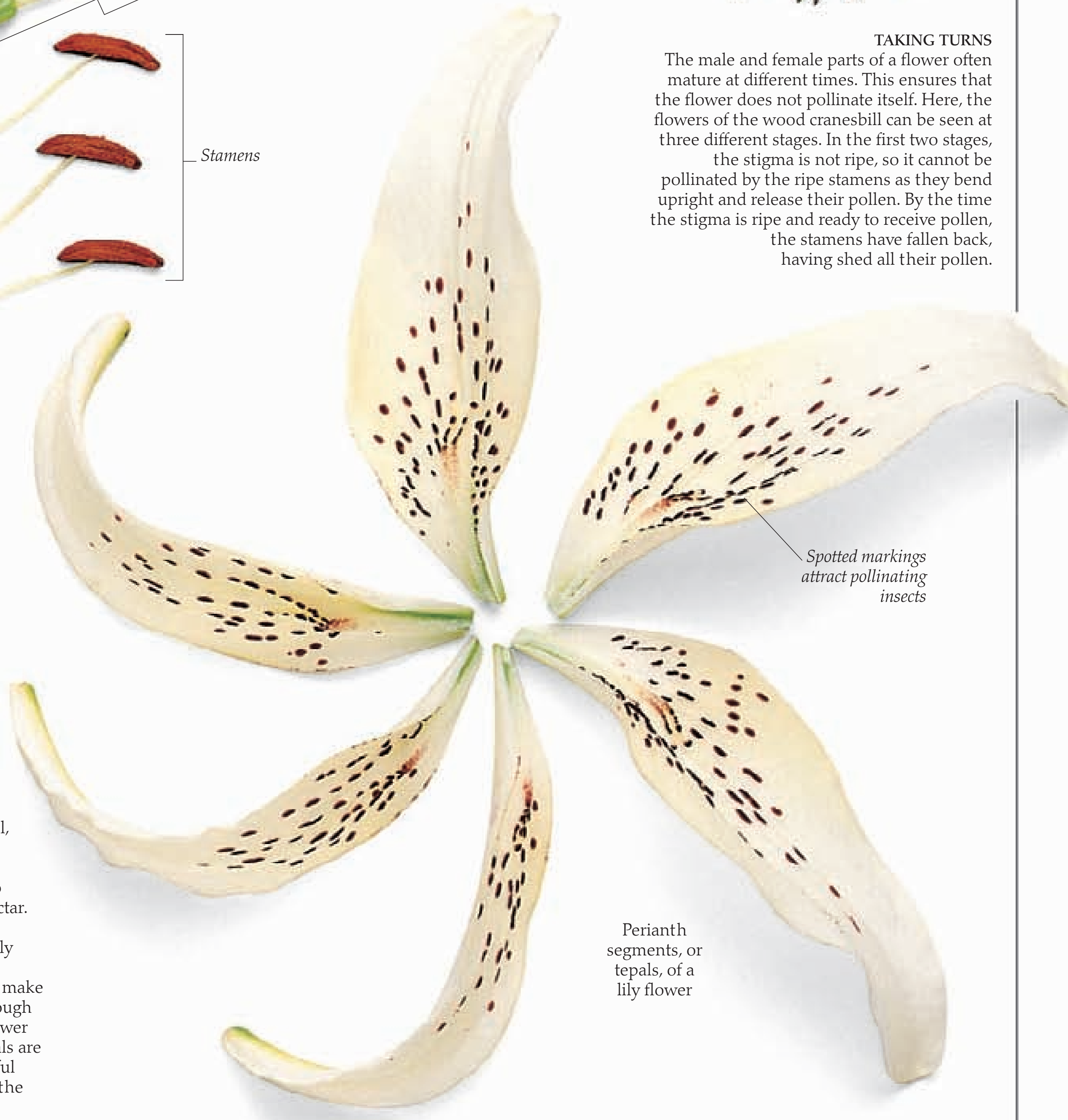
REPRODUCTIVE PARTS

The lily flower contains both male and female parts. The female parts, or carpels, are at the center of the flower. They consist of the ovary, where the seeds are produced, and the stigma, which is attached to the ovary by the style. The stigma is the part of the flower that receives pollen during pollination (pp. 22–23). The male parts of the flower consist of six identical stamens. Each stamen is made up of an anther, which produces the pollen, supported by a filament. As soon as the pollen is ripe, the anthers split open. When an insect visits the flower, some of the pollen brushes off on to its body and will be carried off to pollinate the stigma of another flower.



TAKING TURNS

The male and female parts of a flower often mature at different times. This ensures that the flower does not pollinate itself. Here, the flowers of the wood cranesbill can be seen at three different stages. In the first two stages, the stigma is not ripe, so it cannot be pollinated by the ripe stamens as they bend upright and release their pollen. By the time the stigma is ripe and ready to receive pollen, the stamens have fallen back, having shed all their pollen.



Perianth segments, or tepals, of a lily flower

STAR ATTRACTION

Surrounding the male and female parts of the lily flower is an outer ring, or whorl, of three white sepals, and an inner whorl of three white petals. Because both sepals and petals are identical, they are known as tepals. In the lily flower these tepals work like advertising signs to attract insects in search of nectar. However, not all flowers are like this. In many flowers, only the inner whorl of petals is conspicuous. The sepals that make up the outer whorl may be tough and green and protect the flower bud. In some plants, the sepals are much bigger and more colorful than the petals and perform the task of attracting insects.